Chapter 11

Adaptive Laboratory Evolution for Strain Engineering

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Abstract

Complex phenotypes, such as tolerance to growth inhibitors, are difficult to rationally engineer into industrial model organisms due our poor understanding of their underlying molecular mechanisms. Adaptive evolution circumvents this issue by exploiting the linkage between growth rate and inhibitor resistance to select for mutants with enhanced tolerance. In order to aid experimentalists in the design and execution of adaptive laboratory evolution, we present detailed protocols for batch, continuous, and visualizing evolution in real-time (VERT) approaches to this technique.

Key words: Adaptive evolution, Evolutionary engineering, Complex phenotypes

1. Introduction

Adaptive evolution is a powerful method to improve certain features of common industrial strains (e.g., inhibitor tolerance, substrate utilization, growth temperature) without requiring knowledge of any underlying genetic mechanisms, as long as the desired trait can be coupled with growth. At its core, adaptive evolution simply involves the extended propagation of a microbial strain, typically for hundreds of generations, under the influence of the desired selective pressure. Mutants with enhanced growth rates, both due to increased tolerance or faster consumption of limiting nutrients, will occasionally arise and expand within the population over time. Enhanced strains are therefore obtained by repeated isolation, characterization, and sequencing of these mutant isolates.

Serial batch transfer and continuous bioreactors have been commonly used for evolution experiments. Serial batch transfer has the advantage of being simple to set up and utilize but are more susceptible to random drift due to repeated population bottlenecks. Continuous systems, though requiring a greater initial investment in equipment, avoid this downside by obviating the

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need for repeated inoculations. The relative merits of these distinct experimental setups for different situations are debated below. In addition to how the experiments are carried out, the strain of interest is also a factor. Though any culturable microbial strain can be used in laboratory evolution, the traditional workhorses (*Escherichia coli* and *Saccharomyces cerevisae*) are most often used given their extensive characterization.

Despite the simplicity of laboratory evolution, there are many factors that affect the outcomes of evolution experiments. Factors such as population size, rate of mutation, and frequency of beneficial mutations determine the population structure during evolution and may result in complex population structures resulting from clonal interference, a phenomenon where beneficial mutants compete within a population. Some lineages (individuals descending from specific mutants) are often lost from the population as a result; depending on when mutants are isolated from the population, adaptive mutations present in these mutants may not be identified. If the ultimate goal of the experiment is to use adaptive evolution as a tool to identify the genotypic basis of a particular trait, then the loss of beneficial mutants imposes a limit on the detectable adaptive mechanisms. The visualizing evolution in real time (VERT) pioneered in our laboratory, detailed in this chapter, can be used to observe this phenomenon and assist in the isolation of adaptive mutants. The mutation rate can be enhanced via random whole-cell mutagenesis to increase the genetic diversity of evolving populations so that adaptation can proceed more quickly. Other important variables include the strength of the selective pressure, the population size, and the length of the experiment, all of which will, in combination, influence the evolutionary outcome.

Before the initiation of an evolution experiment, the desired end goal, in terms of phenotypic improvement, should be determined and effective, high-throughput assays for analyzing isolates from the experiment developed. The methods described below are primarily for either *E. coli* or yeast with nutrient limitation as the selective pressure but can be readily modified for any other organism that can be cultured and any other trait that can be coupled with growth.

2. Materials

2.1. Biological and Chemical Materials

1. Target strain for evolution.
2. Defined media (although rich media may be used, defined media is preferred as it allows better control over the condition) in a volume sufficient for the anticipated experimental run.

3. Inhibitor stocks should be sterilized. Antibiotics should be freshly made and stored at −20°C until needed. Fresh supplies of chemical inhibitors should be ordered and stored appropriately.
4. 50% (v/v) glycerol for making frozen stocks of evolving populations.

2.2. Equipment

1. Isothermal incubators, shakers, and water baths.
2. Spectrophotometer capable of absorbance measurements.
3. Microplate reader (e.g., TECAN Infinite) for high-throughput analysis.
4. Optical microscope (e.g., Zeiss Axioscope.A1) for visual inspection of morphology. For yeast, a 40× phase contrast objective (for a total of 400× magnification combined with the 10× magnification of the eye piece) is sufficient. For bacteria, a 100× phase contract objective is needed.
5. Vacuum pumps, lines, and glass bioreactors for continuous experiments.
6. Culture tubes or flasks (sealable if required) for batch experiments.
7. Fluorescent-activated cell sorter (FACS). Commonly used fluorescent proteins such as green fluorescent protein (GFP), yellow fluorescent protein (YFP), and DsRED (RFP) can be excited using a blue laser (excitation wavelength of 488 nm). The exact specifications of the flow cytometer will depend on the properties of the fluorescent proteins used.

3. Methods

We introduce the two main methods for adaptive laboratory evolution: serial batch and continuous systems. General considerations for the design, execution, and analysis of both approaches are presented in the following section. The VERT method, which enables the visualization of competition between different genotypes during evolution, is also detailed. Finally, some additional notes concerning how to use protoplast fusion and mutagenesis to enhance evolutionary outcomes and to prevent contamination are included in Subheading 4.

3.1. Serial Batch Evolution

As mentioned previously, serial batch experiments involve the serial transfer of a population into media containing the inhibitor of interest. Batch experiments may be used if the selection environment does not need to be constant, as culture pH and nutrient composition will change during the course of microbial growth.

External stressors such as nonoptimal temperature may also be used. When evolving for tolerance of volatile chemicals (ethanol, butanol), screw-cap tubes should be used to prevent excessive evaporation during incubation. Be aware that potentially excessive buildup of CO_2 may occur during growth for some organisms. Before the initiation of the experiment, decide whether or not to use mutagenesis to increase genetic diversity (see Note 1). With that caveat in mind, the outline of a general serial batch experiment is given below:

1. Streak out the strain of interest onto agar plates. Grow overnight or until single colonies are visible.
2. Pick one colony for every set of technical replicates and inoculate it into the evolution media. Again, grow the culture until a sufficient cell density is achieved. Replicate cultures from independent colonies (biological replicates) are used to limit the impact of jackpot mutations.
3. Make a glycerol stock of the overnight cultures as the parental strain for subsequent analysis (sequencing, detection of spontaneous mutants, etc.), as mutants may have arisen in the inoculum.
4. Prepare *N* tubes or flasks with *M* ml of media.
   (a) The number of replicates (*N*) should be at least four (two independent colonies as starting parents with two replicates each). Since evolutionary processes are highly stochastic, more replicates are needed if the adaptive evolution experiment is meant to identify distinct molecular mechanisms associated with a trait of interest. If small population sizes are acceptable (small volumes of ~1 ml), 96 replicates can be placed in a 96-well deep-well plates.
   (b) The media volume (*M*) depends on the glassware used in the experiment.
   (c) The inhibitor of interest should already be added to the media, if possible. If not, add it now.
5. Inoculate the replicate cultures with the overnight using 1–2% inoculum (e.g., for a 5 ml culture, use 50 μl of overnight).
6. Incubate at the appropriate evolution condition until the desired benchmark (e.g., cell density, residual substrate) is achieved.
7. During growth, the growth kinetics can be monitored generally via plating, optical density measurements, or cell counting. Measurements for final cell density and other important parameters of interest (e.g., extracellular metabolites, pH of the media, residual nutrients) should be taken before each serial passaging.

(a) A spectrophotometer is best used for a small number of replicates. Microplate readers (e.g., Infinite series from TECAN) are useful for larger (>24) sets).

(b) Keep in mind that optical density is affected by cell size, so mutants with abnormally large or small cell volumes will influence OD readings.

(c) The approximate number of generations can be calculated based on the ratio of initial and final density: Gen = ln $(OD_{final}/OD_{initial})/\ln(2)$.

(d) Prepare another *N* tubes with new evolution media.

8. Perform a tube-to-tube transfer of the inoculum into the new media (replicate 1 to 1). Use a 1–2% inoculum.

   (a) If the cultures are showing signs of growth but are still at a low optical density, allow the overnights to grow for longer (12–24 more hours).

9. Incubate the new replicates until desired benchmark is reached. Keep the overnights at 4°C for one day in case of growth failure, contamination, or other issues; these can be used to establish a new set of replicates if needed.

10. Prepare glycerol stocks of the evolving populations using 67% culture and 33% by volume 50% glycerol. Store at −80°C for later analysis.

11. Repeat steps 7–10 until the desired experimental phenotypic goals have been reached or for a set number of generations or serial passages.

Phenotypic assays can be performed on a routine (e.g., weekly) basis to determine if any desirable mutants have arisen within the population, prior to isolation and detailed characterization. While the initial characterization can be done on a population level to simplify screening, clonal isolates should be used for any sort of detailed analysis. In addition, cell morphology, cell viability, and if relevant, residual media composition (concentrations of residual substrate and/or extracellular products) assays should also be done on a routine basis to detect any unusual changes. Notes 2–4 detail steps that may be taken to avoid contamination of the experiment.

3.2. Continuous Systems

The use of continuous systems, or chemostats, is recommended when a constant physiological state of the microbial system is desired. Continuous systems also offer advantages such as larger population sizes and smaller bottlenecks and allow control of the specific growth rate of the evolving population. The growth kinetics of the culture is dependent on the concentration of the limiting substrate and any growth inhibitors; using a material balance for biomass around the chemostat, the volumetric flow rate (q) over the total volume (V) can be shown to be equal to the net specific growth rate of the culture (see Box 1 for derivation). This ratio of $\mu = q/V$ is called dilution rate (D). Therefore, the average specific growth rate (μ) of the population can be determined by changing the volumetric flow rate (q) of the feed.

Box 1

$$q[x_o] - q[x] + V\mu_g x - Vk_d x = V\frac{dx}{dt}$$
$$[x_o] = 0 : \ \textit{concentration of biomass in feed}$$
$$[x] : \ \textit{concentration of biomass in bioreactor}$$
$$\mu = \mu_g - k_d$$
$$\frac{dx}{dt} = 0 : \ \textit{steady state system}$$
$$\frac{q}{V} = \mu = D$$

3.2.1. Establishing the Limiting Nutrient

Carbon source is generally used as the limiting nutrient in the chemostat, although other essential nutrients (e.g., nitrogen, phosphorous, amino acids for auxotrophic strains) can also be used. If carbon source is the limiting nutrient, keep in mind that glucose-limited cultures seem to be most sensitive to changes in the dilution rate, as lower dilution rates provoke more respiration while higher dilution rates favor fermentation.

1. Inoculate an overnight culture for the strain of interest from a single colony on solid media.
2. Aliquot equal volumes (1–2% inoculum) into a series of appropriate shake flasks that contain different concentrations of the nutrient to be used as the limiting nutrient in the same media (rich or defined) that will be used in the evolution experiments.
3. Determine the growth kinetics by measuring the optical density at different time points at the same temperature that will be used in the evolution experiments.
4. Based on the maximum biomass yield measured from the growth kinetics (optical density at stationary phase), determine the highest concentration of the limiting nutrient that leads to a reduced maximum biomass concentration (earlier onset of deceleration growth phase). Concentration of the substrate lower than or equal to this is limiting.
5. The exact concentration (within the limiting range) of the nutrient is determined by the desired size of the population inside the chemostat. The approximate size of the population

inside the chemostat needs to be measured using different concentrations of the limiting nutrient (within the limiting range) and measured after the system reaches physiological steady state (generally after at least 6–7 volume replacements).

3.2.2. Calculation of Experimental Dilution Rate

Before the start of any continuous cultures, the growth kinetics under the experimental conditions (under the selective pressure to study) must be analyzed. As stated earlier, once the system has achieved steady state, the specific growth rate (μ) of the microorganism corresponds to the dilution rate (D). The dilution rate cannot be set higher than the maximum specific growth rate of the culture, as this will lead to wash out. To determine the maximum specific growth rate, grow the strain under the condition of interest in batch culture with an excess of the limiting nutrient.

1. Streak out the strain of interest on selective solid agar plate to pick a single colony.
2. Inoculate a single colony in 3 ml of selective media at temperature that will be used in the evolution experiments to establish the overnight inoculum.
3. Start batch cultures using 1–2% (v/v) of the overnight inoculum in shake flask or microplate reader. Note: the use of the microplate reader generally underestimates the maximum specific growth rate of the culture.
4. Measure growth kinetics. Calculate the maximum specific growth rate (μ_{max}) of the culture during the exponential growth phase (μ_{max} = slope of the ln (biomass concentration) vs. time plot).
5. Use a μ that is below μ_{max}. The volumetric flow rate (q) can be calculated, since $\mu = D = q/V$, where V represents the volume of the culture. Generally V is a known parameter.

3.2.3. Feed Media

1. Once the volumetric flow rate (q) to be used has been determined, calculate the volume of feed necessary to supply the system for at least 1–2 days (depending on whether the components in the feed are stable and on the ramp-up schedule of the selective pressure, if applicable). It is important to reduce the frequency of feed tank changes to decrease the chances of contamination.

3.2.4. Chemostat Setup

1. Thoroughly clean all the experimental equipment; avoid the use of detergents, as their residues can negatively impact experiments.
2. Assemble the equipment. Sterilize via autoclaving or other sterilization techniques all components that will become in physical contact with feed or cells (see Notes 2–4).

3.2.5. Chemostat Inoculation and Sampling

1. Streak out the strain of interest on selective solid agar plate for single colonies.
2. Inoculate a volume of selective media corresponding to one sixth of V from a single colony and allow growing until stationary phase (overnight) at the experimental conditions. Similar to what was described for serial batch transfers, replicate experiments started from different single colonies are important.
3. Transfer starter culture to chemostats using sterile pipettes using inoculation ports (dependent on the setup of the chemostats).
4. Adjust instruments for experimental conditions (pH meters, thermocouples, flow rates, etc.).
5. Initiate flow rate to start the chemostat.
6. Samples should be taken from the chemostats at least daily. A common sampling regimen includes:
   (a) Prepare glycerol stocks as described in Subheading 3.1. Store at −80°C.
   (b) Inspect cultures under the microscope to determine changes in cell morphology.
   (c) Measure optical density.
   (d) Determine cell viability. Plating on solid media at various dilutions or the LIVE/DEAD® BacLight™ Bacterial Viability Kit (Invitrogen) or propidium iodide staining can be used for this purpose.
   (e) Quantify residual substrate concentration and extracellular metabolite/product concentrations.
7. When it is necessary to change the media, try to do it after sampling to avoid perturbations in the system.

3.3. VERT

3.3.1. Generation of Fluorescent Strains

VERT relies on the use of different fluorescent strains to visualize expansions of different fluorescently marked subpopulations. The adaptive events (changes in proportions of different subpopulations) represent the rise and expansion of adaptive mutants with higher fitness advantage in comparison with the background. VERT requires that the strain to be used for adaptive evolution be marked with different fluorescent proteins. Note that certain fluorescent proteins require oxygen for proper folding; thus, if the evolution were to be carried out under anaerobic conditions, fluorescent proteins that are functional in the absence of oxygen must be used. These fluorescent strains can be generated using classical cloning methods. Some general considerations to take into account when generating fluorescent strains for use with VERT follow:

1. It is best to work with plasmid-free strains to eliminate the need for antibiotics by integrating each fluorescent protein coding sequence into the genome. Integration can be performed using any available technique (Red/ET recombination, CRIM plasmids (1), homologous recombination for yeast, etc.). The use of antibiotics for plasmid maintenance also creates an additional selective pressure for the experiment.
2. Constitutive expression of the fluorescent proteins is preferred (e.g., a protein fusion with a housekeeping gene or use of a constitutive promoter). Chemically inducible promoters may cause extra metabolic burden due to excess expression of the protein, imposing an undesired selective pressure on the strain. This may result in the selection of promoter mutants. In addition, the use of inducers increases experimental cost.
3. It is important to determine whether there are fitness differences between different fluorescently marked strains to ensure an unbiased starting culture. Pairwise competition experiments can be used to determine if such fitness biases exist.

   Day 1

   (a) Inoculate each fluorescent strain from single colonies in evolution media, and grow overnight under the experimental conditions.

   Day 2

   (b) Measure the optical density.
   (c) Prepare different known ratios of each pair of fluorescently marked strains (20:80, 50:50, and 80:20 for every pair of fluorescent strains or just one 50:50 ratio if there are technical limitations) on a cell-density basis. Be sure to have biological (from different starting colonies) and technical (from the same colony, but two inoculations) replicates.
   (d) Measure proportions of the prepared mixtures using FACS. The measured proportions should be close to the known proportions.
   (e) The relative fitness between the two fluorescently marked strains can be determined using either batch cultures or continuous cultures depending on which type of bioreactors the evolution experiment will be carried out. If batch cultures are to be used, prepare 25 ml cultures in baffled flasks of fresh evolution media, inoculating the prepared mixtures in a 1–2% inoculum, and grow overnight under the experimental conditions. If continuous cultures are to be used, then use the mixed culture to seed chemostats under the experimental condition.

Day 3

(f) Measure proportions of the cultures using FACS every 8–12 h (or more frequently if the relative fitness of the two strains are very different).

(g) If batch cultures are used, once the culture reaches mid to late exponential phase, inoculate fresh media using a 1–2% inoculum.

Repeat the suggested protocol to get at least five data points.

(h) If no significant changes (<1% change per generation) are observed in the relative proportions of the two fluorescently marked strains, then they can be assumed to be neutral. If a single replicate shows a consistent bias toward one strain, then it is likely that a jackpot mutation is the cause. However, if significant and consistent differences in the relative fitness of the two strains are observed in all replicates (the same fluorescently marked strain is always expanding against the other one), then these two strains likely do not have equivalent fitness and should not be used together in the evolution experiment if other fluorescent markers can be substituted.

(i) If fitness advantages between the different fluorescent strains are detected, it may be possible to correct the bias by modifying the gene expression. Suggested alternatives include:

1. Modify the extension of the UP elements of used constitutive promoter.
2. Engineering of ribosome binding site (RBS) to tune protein expression.
3. Use alternative constitutive promoters.

3.3.2. Additional Steps

Regardless of whether batch or continuous systems are the selected platform for the evolution experiments, the following steps need to be added to the experimentation when using VERT:

1. *Sampling*. During sampling, it is necessary to measure the proportions of the different fluorescently marked strains using flow cytometer at least daily. More frequent sampling may also be used to provide more granular data.
2. *Isolation of mutants*. VERT facilitates the isolation of adaptive mutants, reducing the size of the population to screen for adaptive mutants. Once the generation that should be sampled has been determined (using the data analysis step described below), do the following:

   (a) Streak out cells from the glycerol stock from the specific generation on solid agar media and incubate until single colonies are visible.

(b) Pick single colonies selecting for the correct fluorescent marker (the color of the expanding subpopulation). When selecting colonies, it is advisable to use a dark reader trans-illuminator instead of a UV lamp to avoid DNA damage. Inoculate colonies in the appropriate media.

(c) Verify the fluorescence of the selected colonies using FACS.

(d) Carry out a pairwise competition as indicated above. Useful comparisons include isolated mutants vs. parental strain and isolated mutant vs. previously isolated mutant. Take into account that the two strains involved in the comparison should express different fluorescent proteins.

(e) An alternative to steps a–b above is to inoculate a large matchstick-sized amount of frozen stock sample from the appropriate generation into the selective media. A FACS sorter can then be used to sort out the desired fluorescently marked subpopulation. Isolation of clones then proceeds as described in steps c and d above to identify the adaptive mutant of interest.

3.3.3. Data Analysis

VERT produces a series of measurements that indicate the relative abundance of cells expressing the different fluorophores (e.g. GFP, YFP, and RFP) within an evolving population. The user can identify frequency changes that correspond to adaptive events manually or computationally using an algorithm developed in our laboratory (2). The latter requires some information concerning variability in FACS measurements, annotated training data denoting adaptive events in real VERT systems, and data from the actual evolution experiments. Example datasets for variability and training calculations are included in the model.

1. Combine the entire series of measurements into one comma-separated text file.
2. Use the first two lines of the file as comments (experiment name and date) and the third as column headers: generation, challenge (inhibitor level), label_1, label_2, label_3.
   (a) Generation refers to the number of generations since the previous measurement.
   (b) Challenge can be fixed at zero if desired.
   (c) The other columns represent frequencies of various fluorophores in the population.
   (d) If only two colors are used, the label_3 column may be omitted.
   (e) Input of data containing four or more colors requires a slight modification of the file input system (see fileRead.m).
3. Place your data in the chemostat_data/experiments folder in the file distribution.

4. Open testVERT.m and change the *filename* variable to match your file.
5. Run the script. The data output will consist of an image with adaptive events' highlight and a line graph of the selective pressure in the system.

The MATLAB package can be downloaded from our laboratory website (http://research.che.tamu.edu/groups/Kao/Web-page/Home.html) or the Journal of Biological Engineering (2), including full instructions on its usage. Octave may also be used to analyze the data without graphical output if desired.

4. Notes

1. Techniques such as protoplast fusion or chemical mutagenesis can be used to speed up adaptation by either inducing recombination between beneficial mutants or increasing the mutation rate. It is best to use mutagenesis sparingly to avoid introducing too many deleterious mutations in the population.
2. In order to detect microbial contamination directly, PCR can be used to amplify strain-specific markers. Sequencing of 16S (prokaryotes) and 18S (eukaryotes) rDNA can also be used.
3. Maintenance of a sterile environment is needed to prevent contamination. For both continuous and batch experiments, all accessories that come into physical contact with the bioreactor must be sterilized. Continuous systems should filter (0.22 μm pore size) inlet air. Media containers should be closely watched for signs of contamination, as foreign organisms will consume nutrients required for growth and may secrete toxins.
4. Sampling from batch and continuous experiments should be done in a sterile environment if possible. In continuous systems, contamination of the effluent tubing is unavoidable but can be reduced by keeping good sterile practices like flaming sampling ports or sterilizing using 70% isopropanol (or 10% bleach) prior to sampling.

References

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